Q8		
(a)(i)	The energy E of a photon is proportional to its frequency ($E = hf$). The atom can only absorb a photon of energy equal to the ***difference*** in two energy levels. From Fig 8.2, we see that the rubidium atom has ***discrete*** energy levels, and thus has discrete energy differences, thus only certain frequencies of photons can be absorbed.	
(a)(ii)	Energy of photon, $\Delta E = hf = \frac{hc}{\lambda} = \frac{(6.63\times10^{-34})(3.00\times10^{8})}{795\times10^{-9}} = 2.502\times10^{-19}\text{ J } (=1.56\text{ eV})$ Energy difference, $E_A - E_{ground} = \Delta E$ $E_A - (-4.18)(1.60\times10^{-19}) = 2.502\times10^{-19}$ $E_A = -4.186\times10^{-19}\text{ J}$ $= -2.62\text{ eV}$	
(a)(iii)	The photon is in the infrared part of the electromagnetic spectrum, outside the visible range, so it cannot be seen with the naked eye. OR The visible spectrum is from 400 nm to 700 nm / 750 nm, thus the photon's wavelength lies outside the visible range. OR The visible spectrum has wavelengths on the order of 10^{-7} m, and the photon's wavelength is near the edge of this range. OR (not intended solution but accepted due to phrasing of the question) A *single* photon reaching the eye has a very small intensity, thus it will be hard to see/detect with the naked eye.	
(b)	$E = \frac{3}{2}kT = \frac{1}{2}mv^2$ $v = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38\times10^{-23})(1000)}{86.9\times1.66\times10^{-27}}}$ $= 535.72 \approx 536\text{ m s}^{-1}$ OR $PV = NkT = \frac{1}{3}Nm\langle c^2\rangle$ $\sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38\times10^{-23})(1000)}{86.9\times1.66\times10^{-27}}}$ $= 535.72 \approx 536\text{ m s}^{-1}$	

(c)	It is the (average) time that the atom stays in that state before it de-excites to another state.	
(d)(i)	Using the de Broglie relation, $p_{photon} = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{795 \times 10^{-9}} = 8.34 \times 10^{-28}$ kg m s^{-1} OR $E = hf = \frac{hc}{\lambda} = pc$ From (a)(ii), $E = 2.502 \times 10^{-19}$ J $\therefore p_{photon} = \frac{E}{c} = \frac{2.502 \times 10^{-19}}{3.00 \times 10^{8}} = 8.34 \times 10^{-28}$ kg m s^{-1} By the principle of conservation of momentum, taking right as positive $p_{atom} - p_{photon} = p'_{atom}$ $\Delta p_{atom} = p'_{atom} - p_{atom} = -p_{photon}$ Thus $\lvert \Delta p_{atom} \rvert = \lvert -(8.34 \times 10^{-28}) \rvert$ $= 8.34 \times 10^{-28}$ kg m s^{-1}	
(d)(ii)	The photons are emitted in randomly in all directions. <u>Since momentum is a vector quantity,</u> the average change in momentum from this process is the <u>vector sum</u> $\Delta \vec{p} = \frac{\Delta \vec{p}_1 + \Delta \vec{p}_2 + ... + \Delta \vec{p}_N}{N} = \frac{0}{N} = 0$	
(d)(iii)	From Table 8, the lifetime of the atom in state A is 27.6 ns. Since from (d)(ii) we know that, on average, only absorbing a photon causes a change in momentum, therefore, $F = \frac{\Delta p}{\Delta t}$ $= \frac{8.34 \times 10^{-28}}{27.6 \times 10^{-9}}$ $= 3.02 \times 10^{-20}$ N	
(d)(iv)	The photons required to excite the atom to state B have shorter wavelength and hence greater momentum, thus <u>each interaction causes a greater decrease in momentum of the atom.</u> OR The lifetime of state B is shorter than state A, so <u>it can absorb more photons per unit time</u> OR <u>the average force on the atom is greater.</u>	
(e)	From Fig. 8.4, when the temperature decreases to T_C, <u>the density of the cloud of atoms increases significantly</u>, suggesting that the particles are overlapping. OR	

	From Fig. 8.4, when the temperature is equal to or below T_C, almost all the particles occupy a small space in the middle of the trap, suggesting that the particles are overlapping.	
(f)	From the de Broglie relation, $\lambda = \dfrac{h}{p} = \dfrac{h}{\sqrt{3mkT}}$ Since a BEC forms when $\lambda \geq d$, where $T \leq T_C$, and $d \approx \dfrac{1}{\sqrt[3]{1.00 \times 10^{19}}} = 4.6416 \times 10^{-7}$ m, $\dfrac{h}{\sqrt{3mkT_C}} = \dfrac{1}{\sqrt[3]{n}}$ $T_C = \dfrac{h^2}{3mk} n^{\frac{2}{3}}$ $= \dfrac{\left(6.63 \times 10^{-34}\right)^2}{3\left(86.9 \times 1.66 \times 10^{-27}\right)\left(1.38 \times 10^{-23}\right)} \left(1.00 \times 10^{19}\right)^{\frac{2}{3}}$ $= 3.4164 \times 10^{-7} \approx 3.42 \times 10^{-7}$ K Alternative solution without using de Broglie relation (max 3 marks only) From the Heisenberg uncertainty principle, $\Delta x \Delta p \gtrsim h$ Taking $\Delta p \approx p = \sqrt{3mkT}$, $\Delta x \gtrsim \dfrac{h}{\sqrt{3mkT}}$ Since a BEC forms when the particles "overlap", i.e. $\Delta x \geq d$, where $T \leq T_C$, and $d \approx \dfrac{1}{\sqrt[3]{1.00 \times 10^{19}}} = 4.6416 \times 10^{-7}$ m, $\dfrac{1}{\sqrt[3]{n}} = \dfrac{h}{\sqrt{3mkT_C}}$ $T_C = \dfrac{h^2}{3mk} n^{\frac{2}{3}}$ $= \dfrac{\left(6.63 \times 10^{-34}\right)^2}{3\left(86.9 \times 1.66 \times 10^{-27}\right)\left(1.38 \times 10^{-23}\right)} \left(1.00 \times 10^{19}\right)^{\frac{2}{3}}$ $= 3.4164 \times 10^{-7} \approx 3.42 \times 10^{-7}$ K	